

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 2

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1. TITLE

Dashboard Design with CSS3 Grid and Flexbox Modules

2. AIM

To design a complex UI utilizing CSS3 Flexbox and Grids, ensuring responsive structures, shadow effects, and hover transitions.

3. OBJECTIVE

The main objectives of this experiment are:

- Abstract styling logic via CSS.
- Align items precisely with Flex.
- Generate column grids.
- Provide rounded corners and drop shadows.
- Trigger animation states on hover.

4. THEORY

Flexbox simplifies distributing space in single dimensions, while Grid excels in complex 2D placements. Together, alongside CSS transitions, they remove the need for JS-based layouts and create highly performant, liquid designs.

5. PROCEDURE

1. Link embedded stylesheet.
2. Set padding and background on root elements.
3. Apply display:flex to navigation block.

4. Construct a `display:grid` system for data cards.
5. Add `border-radius` to card borders.
6. Define `.hover` classes causing scaling on interaction.

6. SOURCE CODE

Source File: `index.html`

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>v2 of week1</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color:rgb(199, 134, 134);
      padding: 4px;

    }
    /* the # is used for unique element here we give main nav at center only for once */
    #main-nav {

      background-color: #f5ff9a;
      padding: 15px;
      border: 8px solid black;
      border-radius: 8px;
      color: rgb(24, 24, 24);
      text-align: center;
      box-shadow: 0PX 4PX 4PX rgb(36, 41, 28);
    }
    /* using . we create a class
    class is usefull dor using the same frame or buttons repeatedly instead of creating
    evrytjhing again and again we can use class*/
    .item-card{
      background-color: rgb(255, 98, 106);
      border:2px solid #92e0da;
      border-radius: 12px;
      padding: 22px;
      margin: 8px;
      box-shadow: 0px 2px 5px #32413f;
      flex: 1;
    }
    /* button is created */
    .btn-main{
      background-color: bisque;
      color: rgb(37, 19, 104);
      padding: 12px 8px;
      border: none;
      border-radius: 5px;
      cursor: pointer;
      box-shadow: 0PX 8PX 12PX rgb(7, 4, 26);
      transition: backgr;
    }
  
```

```

/* I AM MAKING Card for the box at button to getstarted or at the start */
.card{
  width: 200px;
  height: 130px;
  padding: 8px;
  margin: 20px auto;
  background-color: slategray;
  border: 8px ;
  border-radius: 10px;
  font-family : Arial, sans-serif;
  text-align:center;
  box-shadow: 0px 4px 4px black;
}
.container{
  display: flex;
  justify-content: center;
}
.grid-con{
  display: grid;
  grid-template-columns: repeat(2,1fr);
  gap: 20px
  ;
}
.dash-container{
  display: flex;
  justify-content: space-around;
  align-items: center;
  gap: 20px;
  margin-top: 20px;
}
</style>
</head>

      <!-- HERE WE USED CLASSES -->
<body>
  <nav id ="main-nav">
    <h2>
      <strong>IDEAS TO IMPLIMENT</strong>
    </h2>
  </nav>

  <!-- here to this we used whole div and in that div we gave coaitainer which keep the
block as flexible and gaps btw them-->
  <div class="dash-container">
    <div class="item-card">
      <h3>AI</h3>
      <p>Create AI ideas.</p>
      <button class="btn-main">Open</button>
    </div>

```

```
<div class="item-card">
  <h3>Games</h3>
  <p>Design your game.</p>
  <button class="btn-main">Open</button>
</div>

<div class="item-card">
  <h3>Projects</h3>
  <p>Build projects easily.</p>
  <button class="btn-main">Open</button>
</div>
</div>
<!--used classes in the div-->
<div class="grid-con">
  <div class="item-card">
    <h3>TO visualise</h3>
    <p> here u can 3d view ur imagination </p>
    <button class="btn-main">
      visualise
    </button>
  </div>
  <div class="item-card">
    <h3>Upgrades</h3>
    <p>Here U can upgarde ur ideato better</p>
    <button class="btn-main">upgarde</button>
  </div>
</div>

<!-- its a button like to start used class card -->
<div class="card">
  <h2>get started</h2>
  <button class="btn-main">start</button>

</div>

</nav>

</body>
</html>
```

7. SYNTAX EXPLANATION

Syntax Element	Description
border-radius: 10px;	Softens sharp corners.

display: grid;	Invokes matrix layout rules.
align-items: center;	Centralizes vertical flex alignment.
transform: scale(...)	Animates scaling.

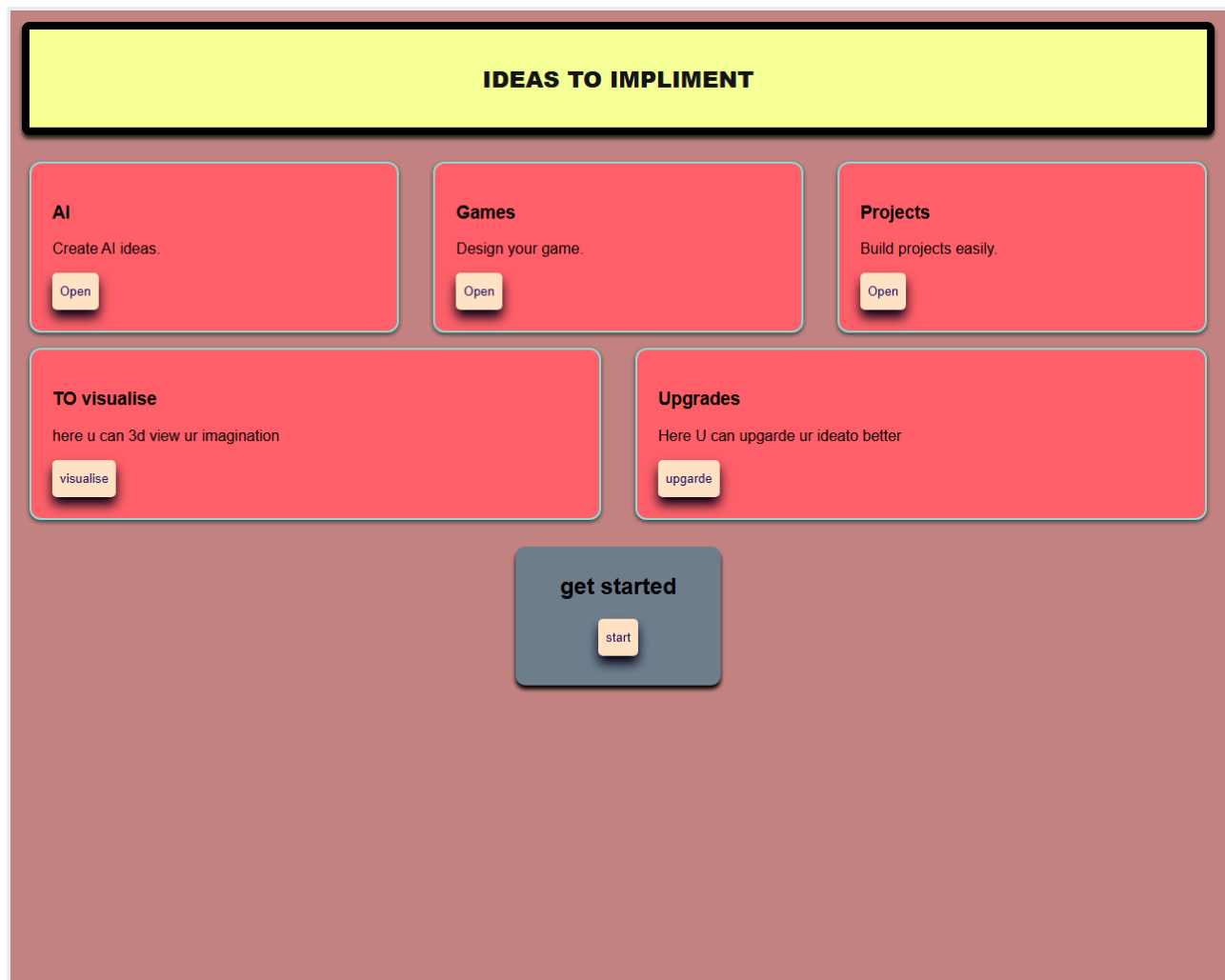
8. TEST CASES & EXECUTION

The following test cases were executed to validate functionality:

ID	Action	Expected Result	Actual Result	Status
T1	Flex Initialization	Align menu items	Menu items aligned	Pass
T2	Grid Construction	Wrap cards neatly	Cards wrapped successfully	Pass
T3	Hover States	Buttons enlarge slightly	Transition verified	Pass
T4	Responsiveness	Elements squeeze on resize	Shrink wrap verified	Pass
T5	Bad Class Name	Add invalid class	Styles fail to mount	Fail

9. PROGRAM OUTPUT

Output A: Grid and Flex Integration Display



10. CONCLUSION

Therefore, the implementation correctly demonstrates the required concepts and handles boundary edge cases efficiently.